## 5.3 Operating conditions

### 5.3.1 General operating conditions

**Table 23. General operating conditions**

| Symbol | Parameter | Conditions | Min | Max | Unit |
|---|---|---|---|---|---|
| $V_{DD}$ | Standard operating voltage | - | 2.0[1] | 3.6 | V |
| $V_{IN}$ | I/O input voltage | - | -0.3 | Min ($V_{DDIO1}$ + 3.6, 5.5)[2] | V |
| $f_{HCLK}$ | AHB clock frequency | - | - | 48 | MHz |
| $f_{PCLK}$ | APB clock frequency | - | - | 48 | MHz |
| $T_A$ | Ambient temperature[3] | Suffix 6[4] | -40 | 85 | °C |
| | | Suffix 7[4] | -40 | 105 | |
| | | Suffix 3[4] | -40 | 125 | |
| $T_J$ | Junction temperature | Suffix 6[4] | -40 | 105 | °C |
| | | Suffix 7[4] | -40 | 125 | |
| | | Suffix 3[4] | -40 | 130 | |

1. When RESET is released, functionality is guaranteed down to $V_{PDR}$.
2. For operation with voltage higher than VDD +0.3 V, the internal pull-up and pull-down resistors must be disabled.
3. The $T_A$(max) applies to $P_D$(max). At $P_D$ < $P_D$(max) the ambient temperature is allowed to go higher than $T_A$(max) provided that the junction temperature $T_J$ does not exceed $T_J$(max). Refer to *Section 6.6: Thermal characteristics*.
4. Temperature range digit in the order code. See *Section 7: Ordering information*.

### 5.3.2 Operating conditions at power-up / power-down

The parameters given in *Table 24* are derived from tests performed under the ambient temperature condition summarized in *Table 23*.

**Table 24. Operating conditions at power-up / power-down**

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| $t_{VDD}$ | $V_{DD}$ rise time rate | 0 | ∞ | µs/V |
| | $V_{DD}$ fall time rate | 10 | ∞ | |

### 5.3.3 Embedded reset and power control block characteristics

The parameters given in *Table 25* are derived from tests performed under the ambient temperature conditions summarized in *Table 23*.

**Table 25. Embedded reset and power control block characteristics**

| Symbol | Parameter | Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| $t_{RSTTEMPO}$[1] | POR temporization when $V_{DD}$ crosses $V_{POR}$ | $V_{DD}$ rising | - | 270 | 500 | µs |
| $V_{POR}$[1] | Power-on reset threshold | - | 1.9 | 1.94 | 1.98 | V |